Rollno:241001036

SOLVE A GRAPH USING DEPTH FIRST SEARCH(DFS)

Python code:

```
def dfs(graph, start, visited=None):
    if visited is None:
        visited = set()

    visited.add(start)
    print(start, end=" ")

    for neighbor in graph[start]:
        if neighbor not in visited:
            dfs(graph, neighbor, visited)




graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
```

```
    'D': [],
    'E': ['F'],
    'F': []
}

print("DFS traversal from node A:")
dfs(graph, 'A')
```

output:

DFS traversal from node A:

A B D E F C

Result:

Thus the program of implementing DFS traversal on a graph to explore all nodes was executed successfully.